MH2500: Probability and Intro to Statistics

Tutorial 10 & Final Exam Preparation

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Law of Total Expectation

Mean of $\mathbb{E}(Y|X)$ is given by $\mathbb{E}[\mathbb{E}(Y|X)]$. Note: $\mathbb{E}(Y) = \mathbb{E}_X[\mathbb{E}_Y(Y|X)]$.

$$\text{Discrete: } \mathbb{E}(Y) = \sum_x \mathbb{E}(Y|X = x)p_X(x).$$

$$\text{Continuous: } \mathbb{E}(Y) = \int_x \mathbb{E}(Y|X = x)f_X(x)\,dx.$$

Corollary (Computing Probabilities by Conditioning)

Given that E is an event:

$$P(E) = \sum_x P(E|X = x)p(x) \text{ or } \int_x P(E|X = x)p(x)dx.$$

For any value of x, we define

$$\mathsf{Var}(Y|X = x) = \mathbb{E}(Y^2|X = x) - [\mathbb{E}(Y|X = x)]^2.$$

Proposition

$$\mathsf{Var}(Y) = \mathsf{Var}[\mathbb{E}(Y|X)] + \mathbb{E}[\mathsf{Var}(Y|X)].$$

Moment Generating Function (MGF)

The moment-generating function (mgf) of a random variable X is

$$M(t) = \mathbb{E}(e^{tX}), \text{ if the expectation is defined.}$$

$$M(t) = \begin{cases} \sum_x e^{tx} p(x), & \text{discrete case.} \\ \int_{\mathbb{R}} e^{tx} p(x)\, dx, & \text{continuous case.} \end{cases}$$

Note: $M(t) = \mathbb{E}(e^{tX})$ may or may not exist for a particular value of t.

Key result of MGF

By differentiating $M(t)$ r times, we have

$$M^{(r)}(0) = \mathbb{E}(X^r).$$

If the MGF exists in a neighbourhood of 0, then so do all the moments and $M^{(r)}(0) = \mathbb{E}(X^r)$.

Essentially $M'(0) = \mathbb{E}(X)$ and $M''(0) = \mathbb{E}(X^2) \Longrightarrow$ useful for Variance.

Theorem

If X has the mgf $M_X(t)$ and $Y = a + bX$, then Y has the mgf

$$M_Y(t) = e^{at} M_X(bt).$$

Theorem

If X and Y are independent random variables with mgf's M_X and M_Y and $Z = X + Y$, then $M_Z(t) = M_X(t)M_Y(t)$ on the common interval where both mgf's exist.

Joint mgf of X and Y:

$$M_{XY}(s,t) = \mathbb{E}(e^{sX+tY}).$$

X and Y are independent if and only if their joint mgf factors into the product of the mgf's of the marginal distributions.

Markov's Inequality

Let X be a *non-negative* random variable, then

$$P\{X \geq t\} \leq \frac{\mathbb{E}(X)}{t} \quad \text{for any } t > 0.$$

Chebyshev's Inequality

Let X be a random variable such that $\mathbb{E}(X)$ and $\mathrm{Var}(X)$ exist. Then

$$P\{|X - \mathbb{E}(X)| \geq t\} \leq \frac{\mathrm{Var}(X)}{t^2} \quad \text{for any } t > 0.$$

If $\mathrm{Var}(X) = 0$, then $P\{X = \mu\} = 1$. (By Chebyshev's Inequality.)

Let $X_1, X_2, \cdots, X_n$, be i.i.d. variables with mean μ and variance σ^2. Let $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \quad S_n = \sum_{i=1}^{n} X_i$. Then,

$$\mathbb{E}(\overline{X}_n) = \mu, \mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n} \text{ and } \mathbb{E}(S_n) = n\mu, \mathrm{Var}(S_n) = n\sigma^2.$$

Law of Large Numbers

Let $X_1, X_2, \cdots, X_i, \cdots$ be a sequence of i.i.d. variables with $\mathbb{E}(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$. Let $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.

Weak Law of Large Numbers: Then, for any $\epsilon > 0$,

$$\lim_{n\to\infty} P\left\{|\overline{X}_n - \mu| > \epsilon\right\} = 0.$$

Strong Law of Large Numbers: $P\left\{\lim_{n\to\infty} \overline{X}_n = \mu\right\} = 1.$

Central Limit Theorem

Let $X_1, X_2, \cdots$ be a sequence of **i.i.d.** random variables having mean μ and variance σ^2. Let $S_n = \sum_{i=1}^{n} X_i$. Then,

$$\lim_{n\to\infty} P\left\{\frac{S_n - n\mu}{\sigma\sqrt{n}} \le x\right\} = \Phi(x), \quad -\infty < x < \infty.$$

Apply continuity correction if X_i happens to be discrete.

Final Exam Information

- Topics:
  - **Basics of Probability & Combinatorics, LOTP, Bayes' Rule.**
  - **Discrete r.v. : Indicator/Bernoulli, Binomial, Poisson, Geometric, Hypergeometric.**
  - **Continuous r.v. : Uniform, Normal, Exponential, Gamma.**
  - **Joint distributions: Marginals, Independence, Conditionals, Ordered statistics, Probability/Expectation of joint pdf/pmf , Convolution.**
  - **Covariance, Correlation coefficient, Conditional Expectation, Expected number of events in n trials.**
  - **Central Limit Theorem and Approximations.**
- Material: Everything except Confidence Intervals and Hypothesis Testing. (MGF is useful / good to know, but not necessary).
- Weightage: **50% of final grade**.
- Date & Time : **2 December 2021, Thursday**, **9 - 11am**.
- Venue: **Exam Halls D & F** (SRC).
- Calculators are **allowed** with **ONE A4 size double-sided helpsheet**.

Consultation Dates and timeslots, Crash Course Revision

Consultation starts from 17th November onwards (likely to be on Zoom)
In general we can start from 9am-5pm (weekdays), but do PM me on telegram first to check if I'm free.

- Slots: (**Green - Available,** **Red - Booked**) Consultation Schedule
- Send me a telegram PM to confirm (30 mins or 1 hour) slots with me.

Crash Course Revision (**29th Novemeber 2021, Monday**) open for everyone (on Zoom), same Zoom link as tutorial.
- Time: ~~**2 - 4.30pm**~~ **1 - 4pm.**
- Agenda: Go thru important things in the course, some selected assignment problems, AY 2020/21 Finals, and some other years PYP questions.

Revisit tutorials, then Compiled Assignment Questions + Review 1-4 to try/retry:
Assign 2 (Q6,Q9), Assign 4 (Q2,Q5), Assign 5 (Q1), Assign 6 (Q1,Q5,Q6), Assign 7 (Q3,Q4,Q5), Assign 8 (Q3,Q5)

Review 1 (Q6,Q11,Q12,Q14,Q15), Review 2 (Q3,Q4,Q5,Q6,Q7)
Review 3 (Q8,Q9,Q10), Review 4 (Q3,4,5(b),6,7,8,10)
2021 Test2 (Q3,Q4,Q5,Q6)

Recommended Past Year Exam Paper Questions:
AY2020/21 Finals - **Everything**. (* Best paper as reference)
AY2019/20 Finals - **Q1, Q3, Q4** (Q5 is not tested, Q2 - Trigonometry)
AY2018/19 Finals - **Q1, Q3, Q4** (Q5 is not tested, Q2 - Convolution)

Bonus (Only if you have done everything):
AY2017/18 Finals - **Q1, Q2, Q3, Q4.** (Exam Paper and its solution will be uploaded together.)

Links to recorded tutorials, reviews and tests

Hover over cursor to access the link.

- Tutorial 1 Password: *Lu#+9f&
- Tutorial 2 Password: m=40L1Yd
- Tutorial 4 Password: f9US*i81
- Tutorial 5 Password: 8w?97Wj1
- Reviews 1 and 2 Password: J=!eg+7S
- Tutorial 6 Password: z#j55rgA
- Tutorial 7 Password: 9ww6s##$
- Tutorial 8 Password: yCik&3Rb
- Review 3 and Solutions of 2020 Test2 Password: r%80#Y8r
- Explanations and comments of 2021 Test2 Password: $1*MB9b7
- Tutorial 9 Password: #KoRkt35
- Tutorial 10 Password: gSz+6@Rh
- Review 4 Password: &F2%YTp0

Take home message

- Do **sanity checks** if possible during the Final Exam, (whatever possible tool you have to solve integrals, check marginal densities)
- Think of conditioning if you are stuck, or splitting into simpler cases.
- Be **clear of the assumptions** used (i.e. independence, mutually exclusive, normally distributed, need for continuity correction, etc.), question yourself if you can use them.
- Familiarize with the standard distributions and indicator random variables, integration of double integrals.
- Lastly, do take time to review part of the content, rather than cramming on a single day.

Thanks For Your Attention!!

All the best for your Final Exams! ☺